

Ex. No.: I

Date: 21/9/24

Calculate Area and Perimeter

Write an Algorithm and draw a Flowchart to Calculate the area and perimeter of a square.

Algorithm:

step 1: start

step 2: Get the length of one side of square as x

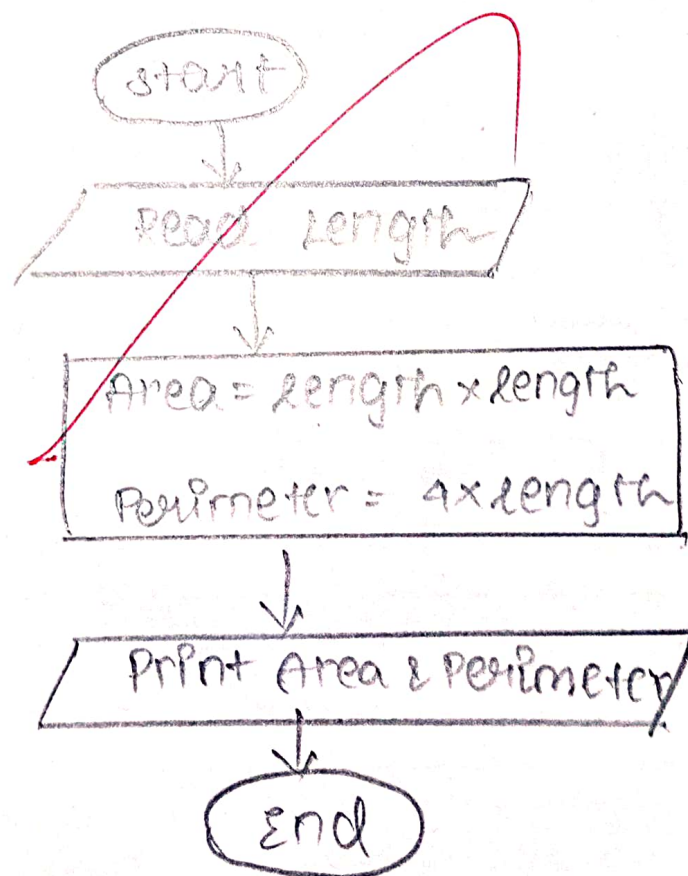
step 3: Area = $x * x$

step 4: perimeter = $4 * x$

step 5: print area & perimeter

step 6: stop

Flowchart:



Ex. No.: II

Date: 21/9/24

Days to Year Conversion

Write an Algorithm and draw a Flowchart to convert the given days into years & months.

Algorithm:

step 1: start

step 2: Get total days

step 3: compute $\text{years} = \text{Total days} / 365$

step 4: compute $\text{Remaining days} = \text{Total days} \% 365$

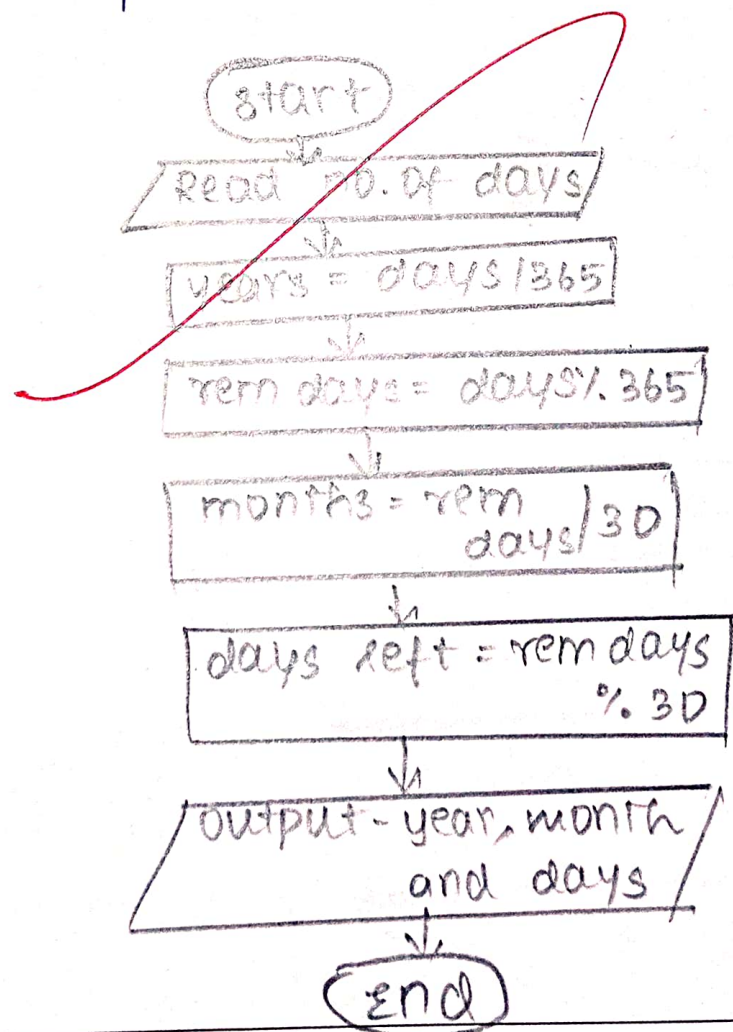
step 5: compute $\text{Months} = \text{Remaining days} / 30$

step 6: compute $\text{days} = \text{Remaining days} \% 30$

step 7: print years, months, days

step 8: stop.

Flowchart:



Ex. No.:

III

Date:

21/9/24

Prime Number

Write an Algorithm and draw a Flowchart to check whether the given number is Prime or not.

Algorithm:

step 1: start

step 2: Get number x

step 3: set count = 0, $i = 0$

step 4: $i = i + 1$

step 5: If $i \leq n$, otherwise go to 8

step 6: If $n \% i = 0$

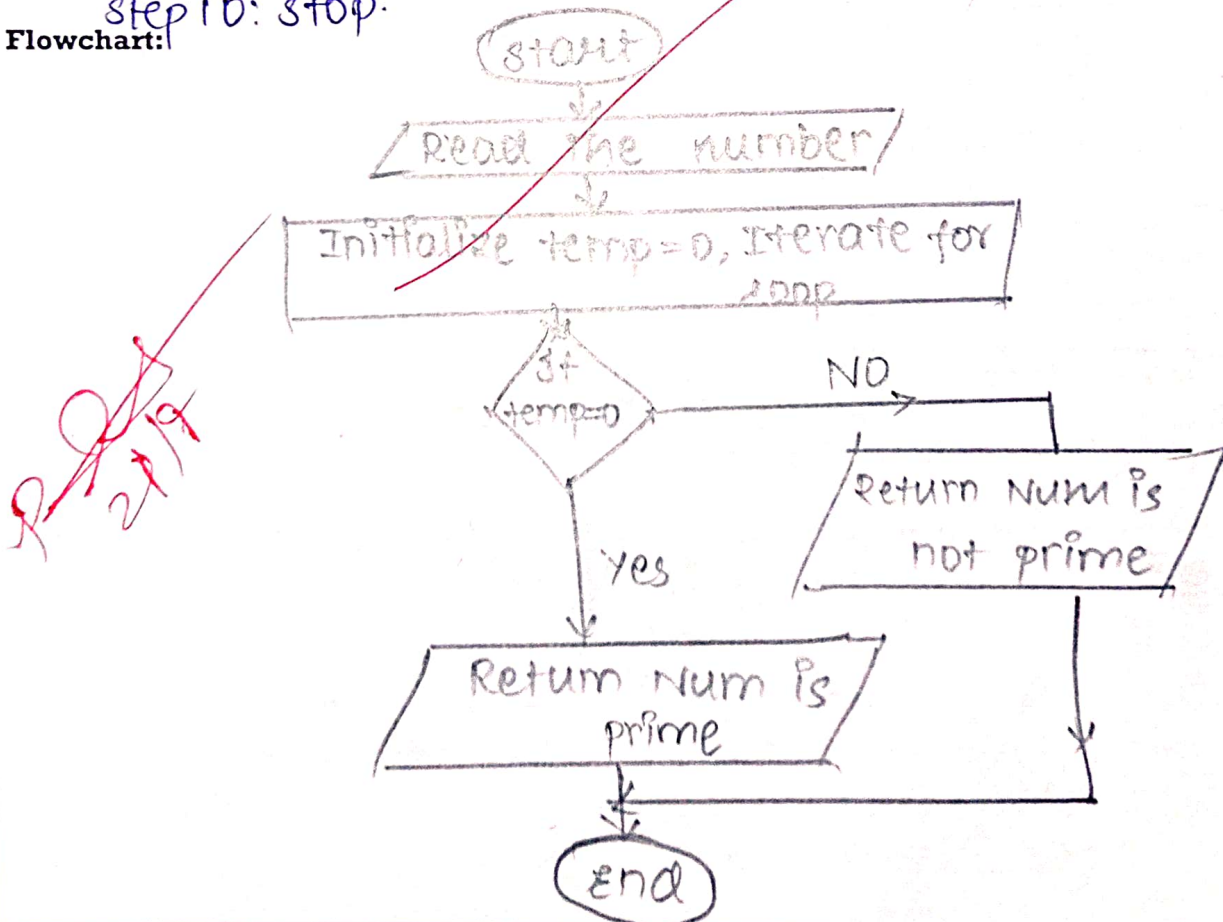
step 7: Increment count by 1, goto 4

step 8: If count = 2, print prime, goto 10

step 9: print n is not prime

step 10: stop.

Flowchart:



Ex. No.: IV

Date: 25/9/24

Leap Year

Write an Algorithm and draw a Flowchart to check whether the given year is Leap year or not.

Algorithm:

step 1: start

step 2: get year

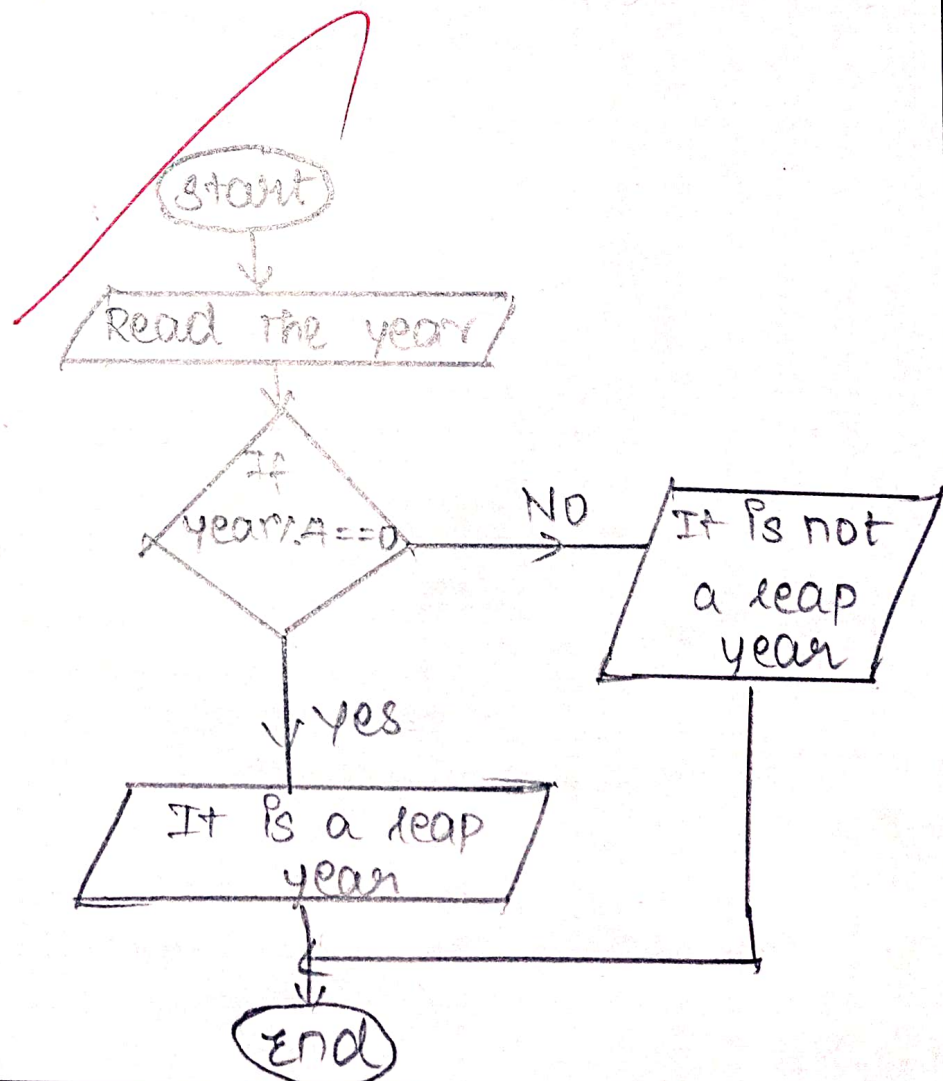
step 3: compute $\text{rem} = \text{year} \% 4$

step 4: If $\text{rem} = 0$, print leap year, otherwise

step 5: print not leap year

step 6: stop.

Flowchart:



Ex. No.: V

Date: 25/9/24

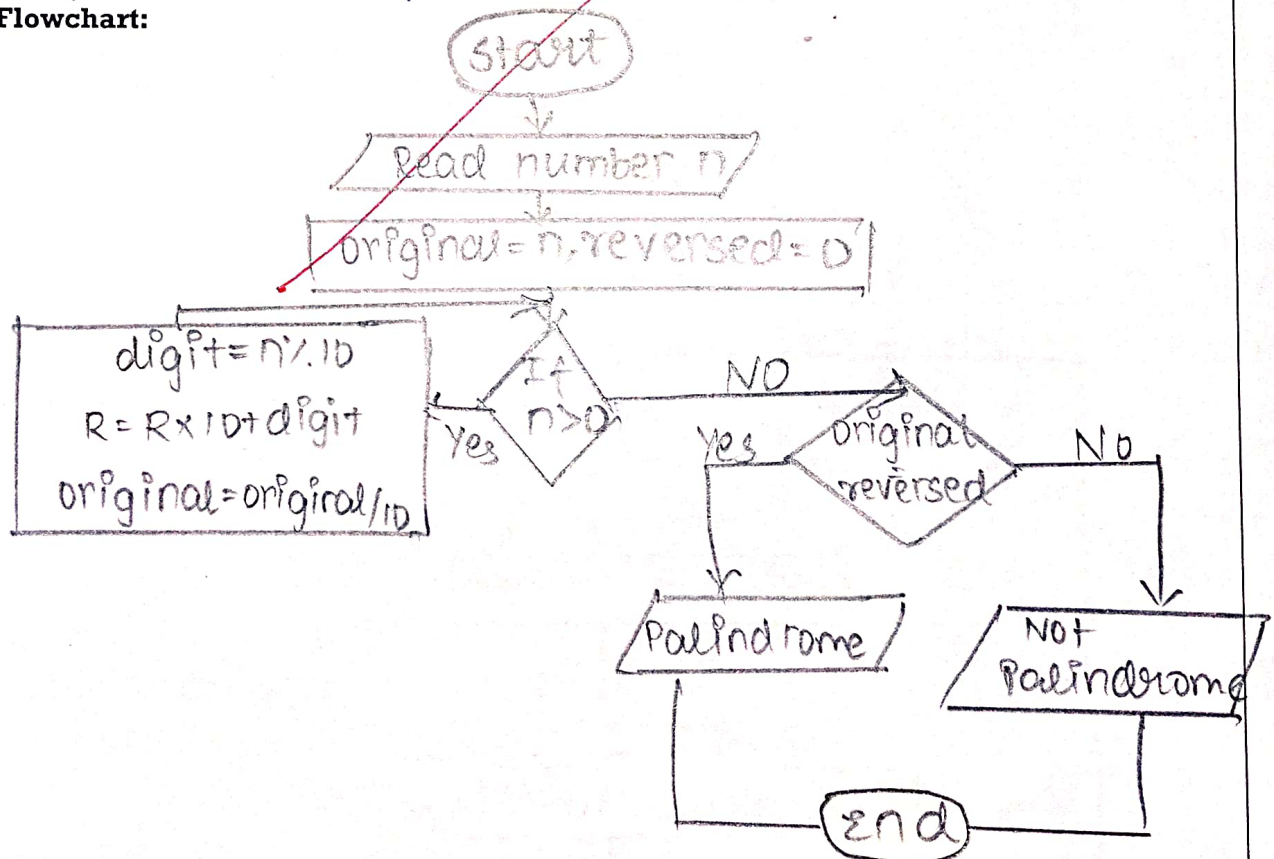
Palindrome Number

Write an Algorithm and draw a Flowchart to check whether the given number is palindrome number or not.

Algorithm:

- step 1: start the process
 step 2: Read the number n
 step 3: Initialize original=n and reversed=0
 step 4: while n>0, set digit = n mod 10, update
 reversed = reversed * 10 + digit, update n = n ÷ 10
 step 5: If original = reversed, print "Palindrome"
 else print "Not palindrome"
 step 6: End the process.

Flowchart:



Ex. No.: VI

Date: 25/9/24

Sum of Digits

Write an Algorithm and draw a Flowchart to calculate the sum of digits in the given number.

Algorithm:

step 1: start

step 2: Read num

step 3: $sum = 0$

step 4: If $num > 0$, otherwise print sum

step 5: $rem = num \% 10$

$sum = sum + rem$

$num = num / 10$

step 6: stop.

Flowchart:

